

Projection-Induced Structure Loss and the Worn-Zhe-Yue Method: A F

****An intuition-guided formalization of claims C011?C025 (all PROVED, novelty: KNOWN, no sorry)***

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****Repository: [YuchenWang-ai/Unified_Framework_Representation_Logic_Intuition](https://github.c**

> ****Statement attribution****: Every statement in this paper about what is **not proved** ? "does n

Abstract

We report a complete formalization, in Lean 4 + mathlib, of a chain of classical results in the

1. Introduction

1.1 Background and motivation

The Riemann zeta function $\zeta(s)$ and its nontrivial zeros constitute one of the most studied obje

This paper does not attempt the impossible ? a proof of the hypothesis. It addresses a methodol

1.2 The intuition chain

The chain was proposed as a sequence of intuitions, each subsequently formalized:

- (i) the complex axis is the projection of "1" in a higher-dimensional structure;
- (ii) primes land on translated integer points;
- (iii) $1/2$ is the symmetry center of the inversion?translation dual;
- (iv) the complex axis is curled (infinity and finiteness become indistinguishable);
- (v) primes are lattice points on a circle; a prime circle, rotated once, pairs.

Each intuition is translated into a precise, checkable statement, labeled with its epistemic st

1.3 Contributions

1. ****A complete Lean formalization**** (C011?C025, all PROVED, no `sorry`) of the infrastructure
2. ****The recoverability theorem**** (§4.7): projection irrecoverably drops structure (the imagina
3. ****The Worn-Zhe-Yue method**** (§6): a proof methodology ? deliberately constructing a projecti
4. ****A critical discussion**** (§7, §8): what the formalization establishes, what it does not, an
5. ****A literature survey and references**** (§9, References).

1.4 Honest boundary (as insisted by the author and DeepSeek)

All results have novelty = KNOWN (restatements of classical mathematics). The Riemann hypothesi

2. Preliminaries: ζ , analytic continuation, and the ComplexAxis framework

2.1 The Riemann zeta function and its analytic continuation

****Definition 2.1**** (Riemann zeta, Euler form). For $\text{Re}(s) > 1$, $\zeta(s) := \sum_{n=1}^{\infty} 1/n^s$. The Euler product $\prod_p (1 - p^{-s})^{-1} = \zeta(s)$,

where the product is over all primes p . The identity follows from the complete multiplicativity

****Theorem 2.2**** (analytic continuation, standard). ζ extends to a meromorphic function on $\mathbb{C}$ wit

In mathlib, the Riemann zeta function is defined (`riemannZeta`), its analytic continuation and

2.2 The ComplexAxis framework

****Definition 2.3**** (higher-dimensional structure). `ComplexAxis := {?a, b? : a, b ? ?}`, with m

****Theorem 2.4**** (C011, the square-root role of J). $J \cdot J = ?1$. In the higher-dimensional structure

****Definition 2.5**** (projection and lifting). $\text{proj } ?a,b? = a$ (drops the rotation component); lif

****Theorem 2.6**** (C011, projection drops structure). proj preserves addition but not multiplication

****Theorem 2.7**** (C011, basepoint drift). The basepoint is $J (= i)$; $\text{proj } i = 0$ (origin illusion)

****Theorem 2.8**** (C011, the real axis is a projection equivalence class). The real-direction line

****Remark 2.9**** (projection as a model of analytic continuation). The projection is a deliberate

3. Result I: the circle structure of primes

This section formalizes the structure of primes as lattice points on circles. The results are

3.1 Sums of two squares

****Theorem 3.1**** (C014, sums of two squares; Fermat, classical; formalized using mathlib's Fermat

Proof. The mathlib theorem ``Nat.Prime.sq_add_sq`` (Fermat's two-square theorem) asserts that f

3.2 The prime circle and its 8 lattice points

****Theorem 3.2**** (C017, unique orbit; formalized via Gaussian-integer UFD). For a prime $p \neq 1$ (m

Proof (narrative). (1) ***Reducibility criterion.*** Let $? = ?a,b?$ with norm p prime. If $? = ? \cdot ?$,

3.3 Structure of the 8 points

****Theorem 3.3**** (C015?C016, orbit structure). The 90° rotation ($\times J$) is a 4-cycle ($R? = \text{id}$), pre

Proof (narrative). The rotation $R(z) = J \cdot z$ has $R? = \text{id}$ because $J? = 1$ ($J^2 = ?1$). It preserves

****Theorem 3.4**** (C023, prime-circle product). The product of the 8 lattice points is $p?$ (4 conj

Proof (narrative). Each conjugate pair $\{z, z?\}$ multiplies to $z \cdot z? = \text{norm } z = p$; the product o

3.4 Splitting and associates

****Theorem 3.5**** (C024, splitting structure). A prime $p \neq 1 \pmod{4}$ splits into a conjugate Gaus

Proof (narrative). The splitting follows from the two-square theorem (3.1) and the UFD: $p = ?$

****Theorem 3.6**** (C020, pairing and intersection). The conjugate pairs are $(a,b)?(a,?b)$, etc., 4

Proof (narrative). The intersection is the pair of points on both circles; for $p = 2$, the cir

4. Result II: curling and critical-line geometry

This section formalizes the "curling" of infinity into finiteness (inversion) and the geometry

4.1 Curling (inversion as a compactness mechanism)

****Theorem 4.1**** (C015, curling). The inversion recip $z = \text{conj } z / |z|^2$ curls infinity back to f

Proof (narrative). Write $z = ?a,b?$. Then $|z|^2 = a^2 + b^2$, and recip $?a,b? = ?a,b?/(a^2+b^2)$. Its

4.2 The critical line and its positional form

****Definition 4.2**** (critical line, C019). The critical line is the set of s with $\text{Re}(s) = 1/2$.

****Theorem 4.3**** (C019, positional form). The critical-line condition $\text{Re}(s) = 1/2 \neq 1 \neq s = \text{conj}$

Proof (narrative). For $s = x + iy$, $\text{Re}(s) = 1/2 \neq x = 1/2$. The condition $1 \neq s = \text{conj } s$ reads

4.3 The critical line is a circle under inversion

****Theorem 4.4**** (C019, critical-line circle). The critical line (vertical line $x = 1/2$) is, und

Proof (narrative). For $z = ?1/2, t?$ on the critical line, recip $z = ?1/2, t? / (1/4 + t^2)$. A

****Theorem 4.5**** (C019, set version). $\text{proj } z = 1/2 \neq \text{recip } z \neq \text{criticalCircle}$ (the critical-line

Proof (narrative, both directions). (i) Forward: from ``critical_line_points``, take t with $z =$

4.4 Symmetry structure of $1/2$

Theorem 4.6 (C013/C018, symmetry as a square). The reflection $s \mapsto 1 - \bar{s}$ centered at $1/2$ is

Proof (narrative). Compute $\varphi(z) = i(iz + (1-i)/2) + (1-i)/2 = z + i(1-i)/2 + (1-i)/2 = z$

4.5 The genuine zero-point view and the recoverability theorem

Theorem 4.7 (C023 ff., zero-point view). After inversion, the 8 points of the prime circle

Theorem 4.8 (C011 ff., recoverability is the formal core of the method). **Lost structure is**

- Not recoverable: i and $-i$ project identically ($\text{proj } 0, 1 = \text{proj } 0, -1 = 0$) is the projection

- Recoverable: the real-axis $\pm$ symmetry is preserved ($\text{proj } (\text{lift } r)) = \text{proj } (\text{lift } r)$; lift

Proof (narrative). (i) Non-recoverability: $\text{proj } 0, 1 = 0 = \text{proj } 0, -1$, and there is no func

5. Result III: Euler product and zero relations

5.1 Convergence of the Euler product

Theorem 5.1 (C025, Euler product). $f(n) = 1/n^s$ is completely multiplicative; for $\text{Re}(s) > 1$
 $\sum_p (1 - p^{-s})^{-1} = \sum_n 1/n^s$.

Proof (narrative). (1) Complete multiplicativity: $1/(mn)^s = (1/m^s)(1/n^s)$. (2) Unique facto

5.2 Identification with mathlib's official ζ

Theorem 5.2 (C025). $\text{riemannZeta } s = \sum_p (1 - p^{-s})^{-1}$ for $\text{Re}(s) > 1$ is mathlib's analytic co

Proof (narrative). The theorem is an equality of functions on the domain of absolute converge

5.3 The zero-free region

Theorem 5.3 (C025). For $\text{Re}(s) > 1$, $\text{riemannZeta } s \neq 0$.

Proof (narrative). On the Euler-product domain, each factor $(1 - p^{-s})^{-1}$ is nonzero (its recip

5.4 Zeros on the circle (conditional verification)

Theorem 5.4 (C025, conditional). $s = 1/2 + it$ is a zero of $\zeta(s)$ iff $t \neq 0$; numerically verified zeros (re

Proof (narrative). This is a conditional statement: if a point s has real part $1/2$ (an extern

6. The Worn-Zhe-Yue method (Worn-Zhe-Yue method)

6.1 The principle

The recoverability theorem (Theorem 4.8) grounds a proof methodology: **by deliberately and prec**

6.2 Components of the method

- Projection construction.** Build a higher-dimensional structure and a chosen projection th
- Structure-loss verification.** Verify that the lost structure is truly non-recoverable (Th
- Fast path.** Show that the retained structure is navigable by the compiled intuition chann
- Honest boundary.** The method organizes and verifies known infrastructure; it does not, by

6.3 Relation to the general framework

The method is a special case of the **projection-induced structure loss** principle in the Unifie

6.4 The author's original words

The author insists on the name **Worn-Zhe-Yue** (Worn-Zhe-Yue; φ = pierce through, ψ = fold, χ = trans

> ???

English rendering of the author's words: "I call it the basepoint-construction-induced Worn-Z

These are the author's own words (his insistence, not an opinion of DeepSeek). The claim, if an
